DATA ENGINEERING REPORT

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By
Roll Number of Student: [2410030027]
Name of Student : JAYA KUMARI
Batch: 2026-27

CANDIDATE'S DECLARATION

I, **JAYA KUMARI** do hereby declare that the internship report titled **"Data Engineering"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Student Name: **JAYA KUMARI**
Roll No:[2410030027]

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature
Student Name: **JAYA KUMARI**
Roll No:[2410030027]

OFFER LETTER

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Ref No: C17Y2026M071452151

Internship Offer Letter

Student Name	: JAYA KUMARI
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: AWS Data Engineering
Internship Duration	: June 2026 - August 2026
AICTE Student ID	: STU69e72d7412bab1776758132

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program.**

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan.**
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments.**
- In the final week, you must appear for a **Final Assessment Test.**
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate.**

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Data Engineering for Data-Driven Organizations	Covers course goals, prerequisites, data-driven decision-making, pipelines, the data engineer's role, modern strategies, and hands-on practice with Amazon S3.
Week 2	Data Elements and Pipeline Design Principles	Covers the five Vs of data, data quality, AWS Well-Architected Framework, modern architectures, pipelines, streaming analytics, and hands-on Athena querying.
Week 3	Securing, Scaling, and Ingesting Data Pipelines	Covers cloud security, scaling infrastructure, ETL vs ELT, data wrangling, cleaning, enriching, validating, and publishing.
Week 4	Efficient Data Management & Big Data	Covers batch vs stream ingestion, ETL with AWS Glue and Kinesis, modern storage pipelines, and Redshift hands-on.
Week 5	Big Data & Preparing Data for Machine Learning	Covers Hadoop, Spark, EMR, and ML lifecycle on AWS including SageMaker, Canvas, modeling, and deployment.
Week 6	Analyzing and Visualizing Data	AWS tool comparison with hands-on analysis using IoT Analytics, QuickSight, and Kinesis Data Firehose.
Week 7	Automating the Pipeline	Automating infrastructure deployment and CI/CD with Step Functions, including ETL pipeline labs using Athena.
Week 8	Bridging to Certification	Overview of AWS certifications, tracks, benefits, and preparation resources.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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Under the Patronage of
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INTERNSHIP COMPLETION CERTIFICATE


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MINISTRY OF EDUCATION

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AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

JAYA KUMARI

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Engineering Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Curriculum Provided by:

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Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4930bef033bb7029d125

Student ID: STU69e72d7412bab1776758132

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Data Engineering is one of the most rapidly evolving fields within Computer Science, concerned with the design, construction, and maintenance of systems and infrastructure that allow for the collection, storage, processing, and delivery of data at scale. With the exponential increase in data generated by modern applications, organisations increasingly rely on skilled data engineers to build reliable pipelines that convert raw data into a usable, analysis-ready form.

This report presents the work undertaken during the Virtual Internship on “Data Engineering”, offered by EduSkills Foundation in association with AICTE (All India Council for Technical Education). The internship provided structured, industry-aligned training covering programming, database management, data warehousing, ETL (Extract, Transform, Load) processes, Big Data frameworks, and cloud-based data services, along with a hands-on project to consolidate the concepts learnt.

4.2 Organization Profile

EduSkills Foundation is a Section 8, not-for-profit organisation that works in association with AICTE to bridge the gap between academia and industry. EduSkills partners with leading technology companies to design and deliver virtual internship programs for engineering students across India, providing them with structured, industry-relevant, hands-on learning experiences free of cost.

Organisation Name: EduSkills Foundation, in association with AICTE

Program Name: Data Engineering Virtual Internship

Curriculum Provided by: AWS Academy

Duration: 8 Weeks (June 2026 to August 2026)

Mode: Virtual / Online, self-paced with mentor support

Institution: IILM University, Greater Noida

4.3 Problem Statement

Organisations today generate massive volumes of data from multiple, heterogeneous sources such as transactional databases, application logs, IoT devices and third-party APIs. Without a properly engineered pipeline, this data remains scattered, inconsistent, and difficult to analyse, which prevents timely and accurate decision-making.

4.4 Project Objectives

- To understand the fundamental concepts, roles, and responsibilities involved in Data Engineering.
- To gain hands-on proficiency in Python and SQL for data extraction, cleaning, and querying.

- To learn the design and implementation of ETL pipelines for structured and unstructured data.
- To understand Big Data processing frameworks such as Hadoop and Apache Spark.
- To explore data warehousing concepts and cloud-based data engineering services.
- To design and build an end-to-end data pipeline as part of a capstone project.

4.5 Scope of the Project

The scope of this internship/project defines the boundaries within which the study, learning, and the capstone pipeline were carried out. It clarifies what has been covered as part of the internship and what has been kept outside its purview, so that the objectives and deliverables remain realistic and achievable within the given duration of 8 weeks.

(a) Functional Scope

- Covers the complete, small-to-medium scale data pipeline lifecycle — from raw data ingestion to final reporting — rather than any single isolated stage.
- Includes collection/ingestion of data from one or more identified source(s) such as CSV files, public datasets, or a sample database.

- Includes data cleaning and pre-processing: handling missing values, duplicate records, incorrect formats, and inconsistent entries using Python (Pandas/NumPy).
- Includes data transformation and application of basic business rules, aggregations, and joins to convert raw data into an analysis-ready form.
- Includes loading of the cleaned and transformed data into a structured database / data warehouse for storage and retrieval.
- Includes basic reporting and visualization of the processed data through summary tables, charts, or a simple dashboard.

(b) Technical Scope

- Restricted to the tools and technologies taught during the internship curriculum — Python, SQL, Hadoop/Spark fundamentals, ETL concepts, and the assigned cloud platform (AWS Academy curriculum).
- Limited to a single-node / sandbox or academic-tier cloud environment, rather than a distributed, multi-node production cluster.
- Data volume used for the capstone project is of a size suitable for academic demonstration and learning, not enterprise-scale big data.

(c) Academic Scope

- Limited to the duration of the internship (8 weeks, June–August 2026) and the curriculum prescribed by EduSkills Foundation, AICTE, and AWS Academy.
- Focused on building conceptual clarity and hands-on skill rather than conducting original research.

(d) Exclusions / Out of Scope

- Production-grade deployment, CI/CD pipelines, and enterprise infrastructure management.
- Real-time / streaming data processing at large enterprise scale (e.g., live Kafka streams handling millions of events per second).
- Advanced machine learning model building, training, or integration — the focus remains on the data engineering pipeline, not on predictive analytics.
- Data security, governance, and compliance frameworks at an organisational level, which are beyond the scope of an academic internship.

These exclusions have been identified as potential directions for future enhancement of the project, beyond the current academic scope.

4.6 Technologies and Tools Used

The following tools and technologies were used and learnt during the course of the internship and the capstone project:

Table 4.1: Technologies and Tools Used

Category	Tools / Technologies
Programming Language	Python (Pandas, NumPy)
Database / Querying	SQL (MySQL / PostgreSQL)
Big Data Framework	Apache Hadoop (HDFS, MapReduce, YARN)
Data Processing Engine	Apache Spark (RDDs, DataFrames, Spark SQL)
ETL & Orchestration	Apache Airflow / [tool used]
Data Warehousing	[e.g., Amazon Redshift / Snowflake / Google BigQuery]
Cloud Platform	[AWS / Microsoft Azure / Google Cloud Platform]
Version Control	Git & GitHub
Visualization	[e.g., Power BI / Tableau / Matplotlib]

4.7 System Architecture

The architecture of the data pipeline developed as part of the capstone project follows a standard multi-layer data engineering flow, consisting of a data source layer, an ingestion layer, a processing/transformation layer, a storage layer, and a presentation/reporting layer, as illustrated below.

Figure 4.1: System Architecture of the Data Pipeline.

Data flows from the source systems into a staging area, where it is cleaned and validated. The transformation layer applies business rules, aggregations, and joins before loading the refined data into the warehouse, from where it is consumed by reporting and visualization tools.

4.8 Methodology

The project was carried out in the following stages:

- Requirement Analysis: Understanding the data sources, format, and reporting requirements.
- Data Collection/Extraction: Gathering raw data from the identified source(s).
- Data Cleaning & Transformation: Handling missing values, duplicates, and inconsistent formats using Python/Spark.
- Data Loading: Loading the transformed data into a structured database/data warehouse.
- Testing & Validation: Verifying data accuracy and pipeline reliability.
- Reporting: Generating summary reports/dashboards from the processed data.

Figure 4.2: Methodology Flow Diagram

4.9 Expected Outcomes

- A clear conceptual and practical understanding of the end-to-end data engineering lifecycle.

- Proficiency in Python and SQL for real-world data manipulation tasks.
- Working knowledge of Big Data tools such as Hadoop and Apache Spark.
- Ability to design and implement a basic ETL pipeline independently.
- A functional capstone project that can be showcased as part of a professional portfolio.
- Improved problem-solving, self-learning, and time-management skills gained through self-paced virtual learning.

4.10 Certificates of Completion and Other Communication/Mail Proof

The Internship Completion Certificate issued by EduSkills Foundation in association with AICTE, under the National Internship Portal and curriculum provided by AWS Academy, is attached below as proof of successful completion of the Data Engineering Virtual Internship.

AICTE-EduSkills Internship Certificate download link

1 message

EduSkills Foundation <noreply@eduskillsfoundation.org>

Fri, 31 Jul, 2026 at 2:13 pm To: JAYA KUMARI <jaya.kumari.cs28@iilm.edu>

EduSkills Foundation

Internship Completion Certificate

Dear **JAYA KUMARI**,

🎉 **Congratulations!**

You have successfully completed **10 weeks AWS Data Engineering** internship with EduSkills Foundation. Your certificate is ready — please log in to your profile and download it.

Steps to Download Your Certificate:

1. Login at: https://aicte-internship.eduskillsfoundation.org/
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Explore Assess2Me

A Small Appreciation Gift for You

As a token of appreciation for successfully completing your internship, we are happy to reward you with **20 FREE Coins** on **Assess2Me**.

Use your reward coins to explore assessments, challenge yourself, and test your skills.

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2. At the top section of the dashboard, click **"Join Assess2Me"**.
3. You will be redirected to **Assess2Me**.
4. Use your **20 FREE Coins** and explore assessments.
5. Test your skills and continue your learning journey.

Keep learning. Keep growing. Keep achieving.

☞ **Visit Assess2Me**

Please post your certificate on your **LinkedIn profile** to get better job opportunities and visibility from corporates.

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1. **World's Largest Professional Platform**

   LinkedIn has **900+ million professionals and 60+ million companies** worldwide.

2. **Recruiters Actively Use LinkedIn**

   Around **8 out of 10 recruiters use LinkedIn** to find and verify candidates.

3. **Profiles Are Checked Before Hiring**

   More than **90% of hiring managers review LinkedIn profiles** before shortlisting candidates.

4. **Better Visibility for Freshers**

   Students who post **certificates, internships, and projects** appear in recruiter searches more often.

5. **Skill-Based Hiring is Growing**

   Employers now focus more on **skills, certifications, and practical experience**.

6. **Higher Chances of Job Opportunities**

   Active LinkedIn profiles with achievements get **more internship and job opportunities**.

🚀 **Tip:** Always **share your certificates, tag EduSkills, mentors, and peers** to increase visibility and professional

📌 **Please Tag the following on LinkedIn:**

@EduSkills Foundation | @AICTE | @AICTE NEAT | @Amazon Web Services (AWS) | @Shyama Rath | @Chandrasekhar Buddha | @Shin Teo | @Pratibha Singh (She/Her) | @Rani Sarojeni Burchmore | @Akanksha Rai Sharma | @IILM University, Greater Noida

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CHAPTER 5: BIBLIOGRAPHY / REFERENCES

[1] EduSkills Foundation, “Data Engineering Virtual Internship Program,” Official Website. [Online]. Available: [insert URL]

[2] AICTE, “AICTE Internship Portal,” Official Website. [Online]. Available: [insert URL]

[3] [Author/Organisation], “[Title of course material/documentation used],” [Publisher/Platform], [Year].

[4] [Add any textbooks, research papers, or online resources referred to during the internship, in IEEE/APA format as prescribed by the department.]